Examiner only

6. Rising energy prices are affecting many households. The government can help households keep their energy bills as low as possible. Domestic buildings account for about 35% of final UK energy consumption. Of this, heating rooms is responsible for about 60% and heating hot water accounts for about 15%. The UK is committed to reduce the CO_2 emissions caused by domestic buildings by 80% by 2050.

Statistics for a typical house with no insulation are shown below. The table also shows government estimates of the current usage of each method.

Part of house	Heat loss from uninsulated house (%)	Method of reducing heat loss	Cost (£)	Annual saving on energy bills (£)	% of homes already using this method
roof	25	loft insulation	450	150	91
windows & doors	10	double glazing	3900	90	70
walls	40	cavity wall insulation	1200	250	42
floor	15	carpets	800	40	90
draughts	10	draught excluders	60	5	85

At present, of the 95% of homes with boilers, 800000 have no controls at all and over 70% lack a timer, sufficient hot water tank insulation, room thermostat and thermostatic radiator valves. It is estimated that upgrading all homes so they have these four features could save 30% of the energy used for heating and hot water.

(a) A heating engineer estimates that to maintain the uninsulated house with no heating controls at 20°C will cost £1800 in energy bills per year.

Use the information above to answer the following questions.

(i) Calculate how much of this energy cost is lost through the roof. [2]

cost = £ ....

(ii) The engineer recommends that loft insulation should be installed first.

Explain why this is the best advice even though it will not make the biggest saving in energy bills. [2]

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Examiner only

(iii) Explain why the engineer can only estimate the cost of the energy bills. [3]

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(iv) After fitting a timer, new hot water tank insulation, room thermostat and thermostatic radiator valves a householder's energy bills would reduce to £1 300. Explain whether this is consistent with the estimated saving of 30% of the energy used. [4]

....

....

(v) Use the equation:

$$\text{units used} = \frac{\text{total cost}}{\text{cost of one unit}}$$

to calculate the decrease in units used if the savings in part (iv) above were made. [3]

Assume each unit costs 18 p.

Decrease in units used = ... kWh

(b) Explain how reducing household energy loss will benefit the environment. [2]

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END OF PAPER